Assignment 2: Model Exploration & Results

Objective

This study aims to implement and evaluate machine learning models using real-world air quality data from Istanbul to support the United Nations Sustainable Development Goals (SDGs). Building upon the initial Exploratory Data Analysis (EDA), this phase of the project focuses on applying a Random Forest Classifier to predict air quality risk levels across various districts. By evaluating key performance metrics and visualizing the model's decision making process, the project seeks to provide a data driven framework for promoting SDG 3 (Good Health and Well being) and SDG 11 (Sustainable Cities and Communities) through predictive environmental monitoring.

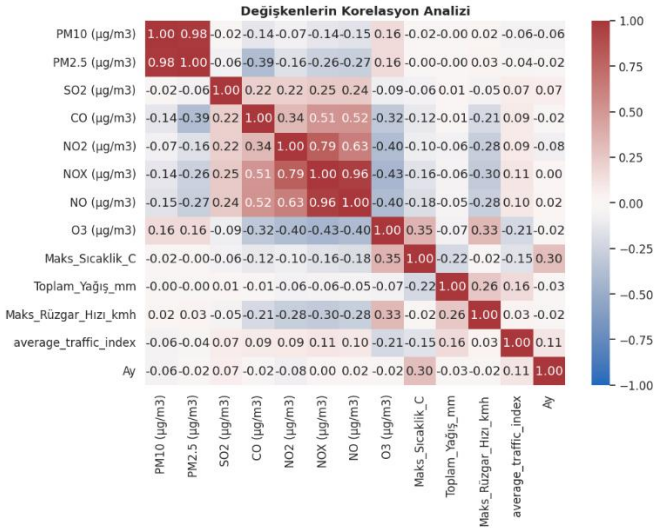
1. Model Selection & Justification

Selected Model: The Random Forest Classifier was selected for this classification task.

Rationale: Random Forest is an ensemble learning method that combines multiple decision trees to produce more accurate and stable predictions. It was chosen because it effectively handles nonlinear relationships between meteorological features and air quality, is robust against outliers, and reduces the risk of overfitting compared to a single decision tree.

SDG Alignment: By predicting air quality risk levels (from Safe to Dangerous), the model provides an early warning system that directly supports SDG 3 (Good Health and Well being) . Additionally, identifying pollution drivers like traffic supports SDG 11 (Sustainable Cities and Communities) by enabling data driven urban planning and environmental management.

2. Data Preparation for Modeling



Data Splitting: The dataset was partitioned into 80% training and 20% testing sets using the `train_test_split` method. The `stratify` parameter was utilized to ensure that the distribution of risk categories remained consistent and representative across both sets.

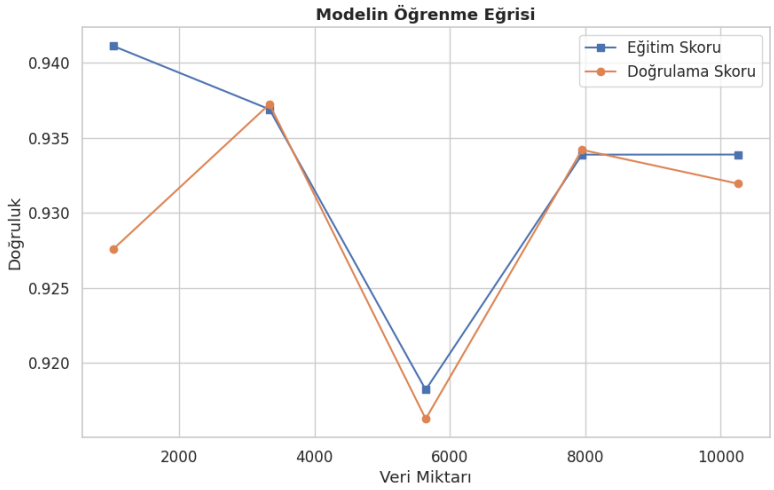
Feature Selection & Engineering: Categorical variables such as 'District' and 'Month' were converted into a numerical format using One - Hot Encoding . This approach allows the model

to interpret geographic and temporal patterns without assuming a false mathematical order between districts.

Feature Scaling: Since meteorological data (Temperature, Wind Speed) and traffic indices have significantly different numerical scales, StandardScaler was applied to normalize the features. This ensures that the model treats all variables fairly during the learning process.

3. Model Training & Evaluation

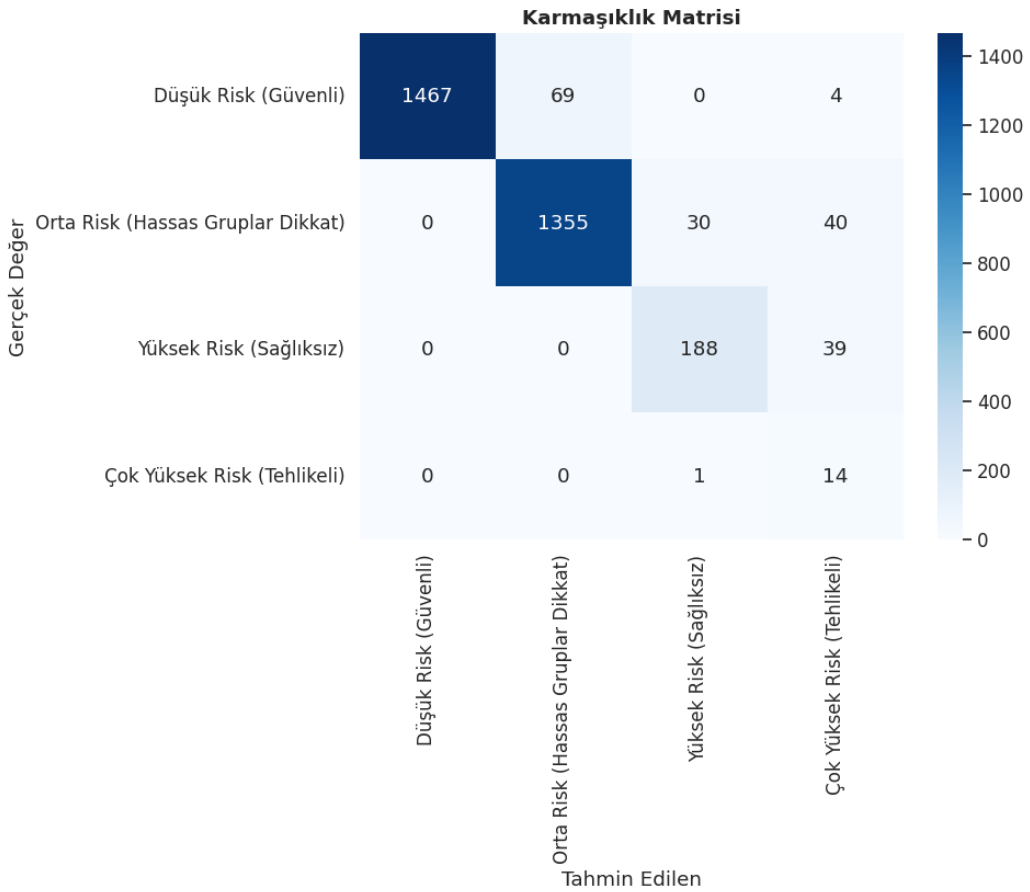
Training: The model was trained using the `rf_model.fit()` function on the scaled training data. To ensure high generalization and prevent overfitting, the `max_depth` was limited to 8, and `class_weight='balanced'` was applied to account for the rare occurrences of high pollution days.



Performance Evaluation: The model achieved an Overall Accuracy of 94.29% and a Weighted F1-Score of 95.20% . These metrics indicate a very high reliability in classification.

Visualizations: A Confusion Matrix was used to evaluate class specific performance. The matrix confirmed that the model has a near perfect success rate in identifying "Low Risk" (Safe) conditions and a high recall for "Very High Risk" (Dangerous) conditions, ensuring that

critical health warnings are captured accurately.



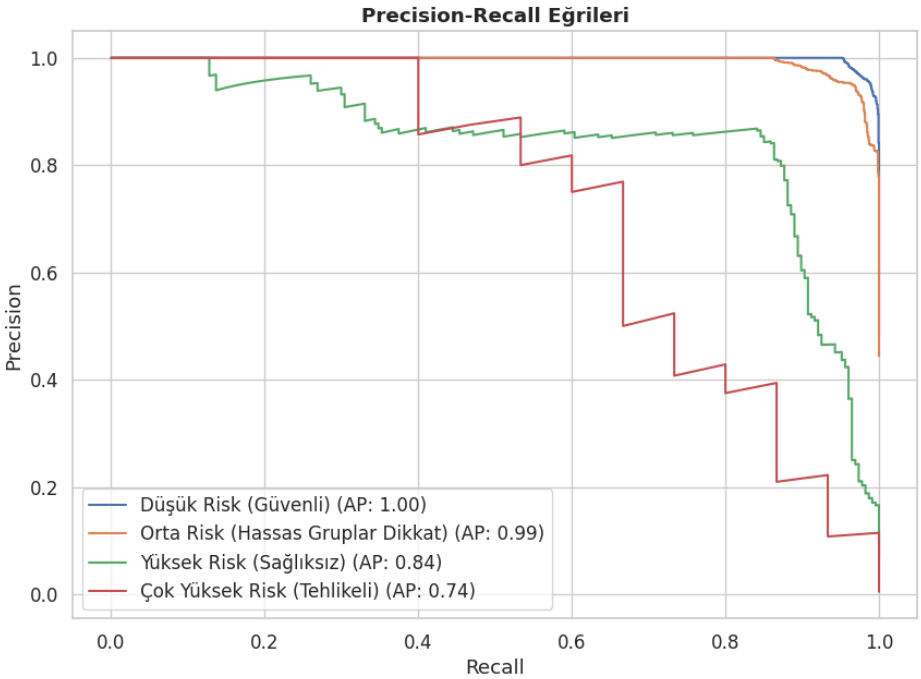
4. Interpretation of Results

Goal Alignment: The model successfully maps environmental and urban data to health risk levels, fulfilling the project goal of creating a predictive tool for Istanbul's district based air quality.

Key Insights: Feature importance analysis revealed that Traffic Index and Temperature are the most significant predictors of air pollution levels. This highlights that urban mobility is a primary driver of air quality fluctuations in Istanbul.

Limitations & Improvements: The primary limitation is the lower Precision for the "Very High Risk" category due to the limited number of extreme pollution samples (imbalanced data). Future improvements could include using SMOTE (Synthetic Minority Over sampling Technique) to generate synthetic samples for rare events or integrating more granular data from

industrial zones.



Team Members

- Mehmetcan Yılmaz - 21061031
- Ibrahim Rzazade - 23061904
- Oğuzhan Ketenci - 20069703
- Zümra Buse Uyur - 21069607